```java
public static int task6CP(int[] input) {
    int n = input.length;                  // Length of the input array
    int maxLength = 0;                     // Variable to store the max length
    int[] dp = new int[n];                 // Array to store lengths of subsequences that end at each index.
    for (int i = 0; i < n; i++) {          // Loop through each element of the input array
        dp[i] = 1;                         // Initialize the length of the subsequence to 1
        for (int j = 0; j < i; j++) {      // Loop through each element before the current element
            if (input[i] > input[j]) {     // If the current element can be included in the increasing subsequence
                dp[i] = Math.max(dp[i], dp[j] + 1);  // Update the length of the subsequence if necessary
            }
        }
    }
    for (int length : dp) {
        maxLength = Math.max(maxLength, length);  // Find the maximum length from the array
    }
    return maxLength;                             // Return the maximum length
}

public static void main(String[] args) {
    int[] input = {10, 9, 2, 5, 3, 7, 101, 18};
    int result = task6CP(input);
    System.out.println(result);
}
```